

Ideation Phase

Brainstorm & Idea Prioritization

Date	29/09/2026
Team ID	SWTID-2026-4654
Project Name	AI StudyBuddy - AI assistant for smarter learning and study support
Maximum Marks	4 Marks

Brainstorming gives every team member room to contribute ideas, with volume valued over polish before the ideas are grouped and ranked.

Step 1: Team Gathering, Collaboration and Select the Problem Statement

The team (leader Siddharth Surya S with members Santhana Barathi R, Subhan Ahamed M, Surendar S and Suresh Babu V) gathered around the case study of **Rahul**, a second-year engineering student whose notes, PDFs, textbooks and articles are scattered and hard to revise. Selected problem statement: *How might we help students turn scattered study material into summaries, flashcards, quizzes and a study plan quickly and securely?*

Step 2: Brainstorm, Idea Listing and Grouping

Group	Ideas listed
Authentication and security	Register and login; JWT access and refresh tokens; bcryptjs password hashing; role-based access for Student and Admin
Study material management	Upload .txt, .md and .pdf files (up to 5 MB) with multer; store title and content; view and manage saved materials
AI generation (Gemini)	Concise summaries; flashcards for active recall; multiple-choice quizzes; personalized study plans from goal, hours per day and days
Storage and history	Save generated resources against the material; retrieve past summaries, flashcards, quizzes and plans
Administration	Manage user accounts; monitor AI service usage; review system logs
Platform quality	Modular MVC structure; global error handler; CORS for the React frontend; health check route

Step 3: Idea Prioritization

Idea	Impact	Effort	Priority
Registration, login and JWT authentication	High	Medium	1 - Must have
Study material upload and storage	High	Medium	1 - Must have
AI summary generation	High	Medium	1 - Must have
AI flashcards and quizzes	High	Medium	1 - Must have
AI personalized study plan	High	Medium	2 - Should have
Saved resources and history	Medium	Low	2 - Should have
Admin user management and monitoring	Medium	Medium	2 - Should have
Advanced usage analytics	Low	High	3 - Could have